

EDUCATION

Heriot-Watt University

Doctor of Philosophy

Edinburgh, UK

10/2023–10/2027 (expected)

Advisors: [Prof. Ahmed H. Elsheikh](#), [Dr. Nanzhe Wang](#)

Research: (i) enhancing the data efficiency of model-based RL through uncertainty estimation and expressive policy classes (diffusion and flow policies); and (ii) applying model-based RL to engineering problems in subsurface flow control

Guangzhou University

Bachelor of Engineering in Robotics Engineering

Guangzhou, China

09/2019–06/2023

SELECTED PUBLICATIONS

- [1] **M. Ye**, A. H. Elsheikh. “DPMPC: Diffusion Policies as Expressive Priors for Model Predictive Control.” *Under review, NeurIPS 2026*.
- [2] **M. Ye**, A. H. Elsheikh. “Model-based reinforcement learning for active flow control.” *Physics of Fluids*, 2025.
- [3] **M. Ye**, J. Zhang. “Mobip: A lightweight model for driving perception using MobileNet.” *Frontiers in Neurorobotics*, 2023.

EXPERIENCE

Doctoral Researcher, Heriot-Watt University, Edinburgh, UK

10/2023–present

- **DPMPC – Diffusion Policy Priors for Model Predictive Control:**

- identified an underexplored bottleneck in high-dimensional MPC — the limited expressiveness of the Gaussian policy prior narrows the diversity of policy-guided proposals used to bootstrap the planner, which constrains exploration and undermines final performance
- developed an efficient recipe for training a diffusion policy prior and incorporated it into the MPC framework (TDMPC)
- achieved substantial improvements over strong baselines across three challenging benchmarks (14 tasks on DMControl, MyoSuite and HumanoidBench), most markedly on humanoid control tasks with 61-dimensional action spaces

- **Model-Based RL for Active Flow Control:**

- designed two control problems (state space, reward function, etc.) from scratch — synthetic-jet drag reduction on a cylinder and subsurface reservoir production maximization
- adapted two model-based algorithms, planning (PETS) and Dyna-style RL (MBPO), to solve the flow control problems; improved the world model with a multi-step predictive loss, alleviating the compounding-error problem during autoregressive rollout
- both MBRL algorithms outperformed the PPO baseline with $2\text{--}9\times$ better sample efficiency and $2\text{--}7\times$ faster convergence in wall-clock time; released code in the Gym-preCICE ecosystem

Visiting Researcher, Tongji University, Shanghai, China

09/2025–11/2025

- **Reduced-order world models for subsurface flow control:**

- used proper orthogonal decomposition (POD) to compress the high-dimensional flow state (pressure and velocity fields) into a compact latent representation, and learned the dynamics model directly in that latent space; the latent model was faster to roll out than the full-order model, enabling real-time planning and control

Visiting Researcher, NORCE Research, Bergen, Norway

07/2025–08/2025

• **Geosteering trajectory control under subsurface uncertainty:**

- formulated real-time geosteering as a partially observable control problem: the drilling trajectory must stay inside the reservoir while the geology ahead of the drill bit is unknown and only indirect, noisy gamma-ray logs are observed
- adapted a world model to predict the gamma-ray response of the surrounding formation, attaining a probabilistic forecast model of the subsurface; coupled that forecast with uncertainty-aware model-based planning to reduce the risk of steering out of the reservoir

Research Assistant, Guangzhou University, Guangzhou, China

03/2022–09/2023

• **MobiP — Lightweight Multi-Task Driving Perception:**

- built a lightweight multi-task perception network for autonomous driving that runs on embedded hardware rather than a dedicated GPU, widening where such algorithms can be deployed
- designed an architecture with a shared MobileNetV2 encoder and two-decoder heads — one detection head for traffic objects and one multi-class segmentation head covering drivable area and lane lines
- with 7.5M parameters, MobiP matched the performance of YOLOP (the SOTA method) while running 45% faster, reaching 58 FPS on an NVIDIA Tesla V100 and 4.8 FPS on a Raspberry Pi 4B

HONORS & AWARDS

- James Watt Scholarship (fully-funded PhD), Heriot-Watt University 2023
- Second-Class Scholarship, Guangzhou University (top 5% of students), awarded twice 2020, 2023
- Third Prize, 12th Chinese National College Students Mathematics Competition 2020

INVITED TALKS

- “Model-Based Reinforcement Learning for Efficient Subsurface Flow Control,” NORCE Research, Bergen, Norway 08/2025

TECHNICAL SKILLS

Programming: Python, C, MATLAB, Simulink, $\text{\LaTeX}$, HTML/CSS

ML: PyTorch, JAX, OpenCV

Robotics: ABB IRB 1600, Arduino, Raspberry Pi, STM32, MuJoCo, Gazebo

Software: SLURM, Docker

Languages: Chinese (Native), English